

Distance moved from initial position at M to second minimum intensity = $3(1/4 \lambda) = 0.30 \text{ m}$

19 A $ke = \frac{1}{2}mv^2 \propto v^2$

using $v^2 = 0 + 2ax \Rightarrow v^2 \propto x \Rightarrow v \propto \sqrt{x}$ and $ke \propto x$, so quantity Y is speed v

E field is uniform inside the parallel plates, electric force is constant.

Electric potential energy, $U = eV$ decreases linearly with x since e is negative and V is increasing linearly from $-V$ to 0 .

20 C